1. Lab 09: IPv4, DHCP, and NAT

Lab Name: Lab 09: IPv4, DHCP, and NAT
Date: YYYY-MM-DD
Name: [Your Name]
Lab Partners: [Partners' Names, if any]
Software/Hardware Used: Cisco Modeling Labs (CML) - Personal / Free Edition

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AI Policy & Acceptable Use Guidelines

 AI Policy & Acceptable Use Guidelines - Yellow Icon

Artificial Intelligence tools can be powerful learning aids, but they must be used ethically and responsibly to ensure you develop core networking skills.

- **Lab Assignments:** Do not use AI to perform lab steps or write your lab submissions. However, if you get stuck on a concept or technical obstacle, you may use AI as a virtual tutor to ask clarifying questions and help you troubleshoot.

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2. Background

Dynamic Host Configuration Protocol (**DHCP**) and Network Address Translation (**NAT**) are foundational technologies for managing IPv4 addressing in modern networks:

1. **Dynamic Host Configuration Protocol (DHCP — RFC 2131)**:
   - Automates the assignment of IPv4 addresses, subnet masks, default gateways, and DNS servers to client hosts using the 4-step **DORA** handshake (**D**iscover, **O**ffer, **R**equest, **A**cknowledge) over UDP ports **67** (server) and **68** (client).
2. **Network Address Translation / Port Address Translation (NAT Overload / PAT — RFC 3022)**:
   - Allows multiple hosts on a private internal LAN (e.g., `192.168.1.0/24`) to share a single globally routable public IPv4 address (e.g., `203.0.113.1`) when communicating with external networks.
   - As packets traverse the router from the `ip nat inside` interface to the `ip nat outside` interface, the router rewrites the private source IP address (`Inside Local`) to its public outside IP address (`Inside Global`) and tracks each connection in its NAT translation table using unique Layer 4 port numbers or ICMP identifiers.

In this lab, you will configure a Cisco IOSv router as both a DHCP server and a NAT Overload gateway in CML, capture the DHCP 4-way handshake and NAT packet translation in Wireshark, and analyze how headers change across the inside and outside links.

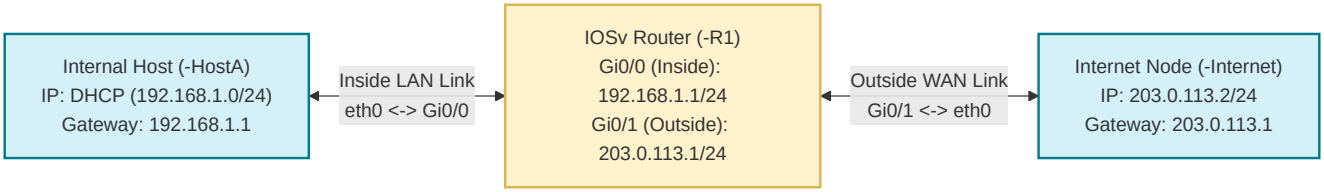
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3. Objective / Purpose

To configure a Cisco IOS router as a DHCP server for internal LAN clients, implement Port Address Translation (NAT Overload / PAT) to allow private IPv4 hosts to reach an external network, and capture and analyze both DHCP DORA negotiation and NAT header translation using Wireshark `.pcap` traces.

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4. Topology Diagram



Topology Details:

- **1x Cisco IOSv Router (`<initials>-R1`)**: Acts as the Inside LAN Default Gateway, DHCP Server, and NAT Overload Gateway.
- **1x Internal Linux Node (`<initials>-HostA`)**: Connected to R1 `Gi0/0` (`Inside LAN`). Obtains its IPv4 address dynamically via DHCP.
- **1x External Linux Node (`<initials>-Internet`)**: Connected to R1 `Gi0/1` (`Outside WAN`). Configured with static public IP `203.0.113.2/24` .

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator, Wireshark
- **Hardware / Virtual Nodes:** 1x Cisco IOSv Router, 2x Alpine Linux Nodes

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6. Step-by-Step Configuration & Packet Capture Procedure

IP Addressing Table

Device	CML Node Name	Interface	IP Address	Subnet Mask	Default Gateway	NAT Role
Internal Host	`<initials>-HostA`	`eth0`	Dynamic (`192.168.1.x`)	`255.255.255.0` (`/24`)	`192.168.1.1` (via DHCP)	Inside Client
NAT Router	`<initials>-R1`	`Gi0/0` (Inside)	`192.168.1.1`	`255.255.255.0` (`/24`)	N/A	`ip nat inside`
NAT Router	`<initials>-R1`	`Gi0/1` (Outside)	`203.0.113.1`	`255.255.255.0` (`/24`)	N/A	`ip nat outside`
Internet Node	`<initials>-Internet`	`eth0`	`203.0.113.2` (Static)	`255.255.255.0` (`/24`)	`203.0.113.1`	Outside Host

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Step 1: Configure Router R1 (`<initials>-R1`)

Open the **Console** tab on Router R1 and apply the interface, DHCP, and NAT Overload configurations (replace `KH` with your own initials):

```
! 1. Enter Global Configuration Mode and set hostname
enable
configure terminal
hostname KH-R1
ip routing

! 2. Configure Inside LAN Interface (Gi0/0 facing Internal HostA)
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 ip nat inside
 no shutdown
 exit

! 3. Configure Outside WAN Interface (Gi0/1 facing Internet Node)
interface GigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.0
 ip nat outside
 no shutdown
 exit

! 4. Configure DHCP Server Pool for the Inside LAN (192.168.1.0/24)
! Exclude the router's own gateway IP (192.168.1.1) so it is not leased to a client
ip dhcp excluded-address 192.168.1.1

ip dhcp pool LAN
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 192.168.1.1
 exit

! 5. Configure NAT Overload (PAT) from Inside LAN to Outside Interface Gi0/1
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/1 overload

! 6. Save configuration
end
write memory
```

What These Router Commands Do:

- **`ip nat inside` / `ip nat outside`** : Marks `Gi0/0` as the private internal network interface and `Gi0/1` as the public external network interface. NAT translation is triggered only when packets are routed between an `inside` interface and an `outside` interface.
- **`ip dhcp excluded-address 192.168.1.1`** : Reserves the router's own gateway IP address (`192.168.1.1`) so the DHCP service starts assigning client leases at `192.168.1.2` .
- **`ip dhcp pool LAN`** : Defines the `192.168.1.0/24` address pool, advertising `192.168.1.1` as the client's Default Gateway (`default-router`) and DNS Server (`dns-server`).
- **`access-list 1 permit 192.168.1.0 0.0.0.255`** : Standard Access Control List (ACL 1) using wildcard mask `0.0.0.255` to match any source IP in the `192.168.1.0/24` subnet.
- **`ip nat inside source list 1 interface GigabitEthernet0/1 overload`** : Instructs R1 to translate the source IP of any packet matching ACL 1 to the public IP address assigned to `GigabitEthernet0/1`

(`203.0.113.1`). The **`overload`** keyword enables Port Address Translation (PAT) so multiple internal hosts can share that single outside IP concurrently.

Step 2: Configure the External Internet Node (`<initials>-Internet`)

Open the **Console** tab on `<initials>-Internet` and configure its static public IP address (`203.0.113.2/24`) and default gateway (`203.0.113.1`):

```
# On Internet Node Console (Alpine Linux):
sudo ip addr add 203.0.113.2/24 dev eth0
sudo ip link set eth0 up
sudo ip route add default via 203.0.113.1
```

Verify connectivity from the Internet node to R1's outside interface by running `ping -c 2 203.0.113.1` .

Step 3: How and When to Start Packet Captures & Generate Traffic

> *[!IMPORTANT]* ***REMINDER: All screenshots, console outputs, and packet captures (.pcap files) must be taken from the SAME active lab session. Always save and submit packet capture files strictly in `.pcap` format (do NOT use `.pcapng`).***

To capture **both** the DHCP 4-way DORA handshake (which occurs only on the Inside LAN link) and the NAT-translated packets (which appear with the rewritten public IP on the Outside WAN link), follow this exact sequence:

1. Start Packet Captures in CML (BEFORE Requesting DHCP or Pinging)

1. **Start Inside LAN Capture (for DHCP DORA & Pre-NAT Traffic):**
   - In the CML workspace, right-click (or click) the **Inside LAN link** between `<initials>-HostA` (`eth0`) and `<initials>-R1` (`Gi0/0`).
   - Open the **Packet Capture** tab at the bottom pane, leave the packet filter blank (or set `bootp or icmp`), and click **Start**.
2. **Start Outside WAN Capture (for Post-NAT Translated Traffic):**
   - Right-click (or click) the **Outside WAN link** between `<initials>-R1` (`Gi0/1`) and `<initials>-Internet` (`eth0`).
   - Open the **Packet Capture** tab, leave the packet filter blank (or set `icmp`), and click **Start**.

2. Trigger the DHCP 4-Way DORA Handshake on `<initials>-HostA`

With the **Inside LAN Capture actively running**, open the **Console** tab on `<initials>-HostA` and request a dynamic IPv4 lease from Router R1:

```
# On Internal HostA Console (Alpine Linux):
sudo ip link set eth0 up
sudo udhcpc -i eth0
```

What happens on the wire: `udhcpc` broadcasts a **DHCP Discover** (`0.0.0.0` -> `255.255.255.255`), R1 responds with a **DHCP Offer** (`192.168.1.2`), HostA sends a **DHCP Request**, and R1 finalizes the lease with a **DHCP ACK**.

- Verify that `<initials>-HostA` received its IP (`192.168.1.2/24`) and default gateway (`192.168.1.1`):

```
ip addr show eth0
ip route
```

- *(Troubleshooting Tip: If `HostA` already had an IP address before you started the capture, run `sudo udhcpc -R -n -i eth0` or `sudo ip addr flush dev eth0 && sudo udhcpc -i eth0` while the capture is running to force a fresh DORA exchange).*

3. Trigger NAT Overload Translation Across Router R1

While **both** packet captures are still running, ping the external Internet Node (`203.0.113.2`) from `<initials>-HostA` :

```
# On Internal HostA Console:
ping -c 4 203.0.113.2
```

What happens on the wire:

- On the **Inside LAN Link (`HostA <-> R1 Gi0/0`)**, the ICMP Echo Request has Source IP `192.168.1.2` and Destination IP `203.0.113.2` .
- On the **Outside WAN Link (`R1 Gi0/1 <-> Internet`)**, Router R1 rewrites the Source IP to `203.0.113.1` (`Inside Global`) and forwards the packet to `203.0.113.2` . When `203.0.113.2` replies to `203.0.113.1` , R1 looks up the connection in its NAT table and translates the destination IP back to `192.168.1.2` !

4. Collect Router CLI Verification Outputs (Immediately After Pinging!)

> *[!WARNING] **Timing Note for `show ip nat translations` :** ICMP NAT translation entries in Cisco IOS time out after **60 seconds** of inactivity! Run `show ip nat translations` on R1 **immediately** after running `ping -c 4 203.0.113.2` on HostA so the active ICMP translation rows are visible in your screenshot.*

On Router `<initials>-R1` , execute and screenshot both commands:

```
show ip dhcp binding
show ip nat translations
```

- **`show ip dhcp binding`** : Confirms the leased IP (`192.168.1.2`) and Client-ID/Hardware MAC address of `HostA` .
- **`show ip nat translations`** : Displays `Pro` (`icmp`), `Inside global` (`203.0.113.1:<id>`), `Inside local` (`192.168.1.2:<id>`), `Outside local` (`203.0.113.2:<id>`), and `Outside global` (`203.0.113.2:<id>`).

5. Stop and Download the `.pcap` Capture Files

1. Click on the **Inside LAN link**, go to **Packet Capture**, click **Stop**, and click **Download**.
2. Click on the **Outside WAN link**, go to **Packet Capture**, click **Stop**, and click **Download**.
3. **File Format Requirement (`.pcap` ONLY — NO `.pcapng`)**:
    - CML downloads files natively in standard `.pcap` format.
    - You may either upload both raw `.pcap` files (`inside_dhcp.pcap` and `outside_nat.pcap`) **or** merge them in Wireshark (`File -> Merge...`) into a single `dhcp_nat_traffic.pcap` file.
    - **Important:** If you save or export from Wireshark, you **MUST** select **`Wireshark/tcpdump/... - pcap (*.pcap)`** in the file type dropdown (do **not** save as the default `.pcapng` , as Canvas only accepts `.pcap` uploads).

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7. Wireshark Analysis & Written Questions

Open your downloaded `.pcap` capture(s) in Wireshark and answer the following questions in your PDF report:

Part 1: DHCP Protocol Analysis (4-Way DORA Process)

1. Filter your Inside LAN capture using `bootp` (or `dhcp`). Locate the 4 messages of the **DHCP DORA** sequence (**Discover**, **Offer**, **Request**, **ACK**):
   - Detail each of the 4 steps in order.
   - For each message, identify the **Source IP / Destination IP** and **Layer 2 Destination MAC address** (Broadcast `ff:ff:ff:ff:ff:ff` vs. Unicast), and explain *why* broadcast or unicast addressing is used at each stage before the client's IP stack is initialized.
   - Expand the **Dynamic Host Configuration Protocol** payload in the `DHCP Offer` / `DHCP ACK` packet: which DHCP Options carry the **Subnet Mask**, **Router (Default Gateway)**, **DNS Server**, and **IP Address Lease Time**?
   - Include an **annotated Wireshark screenshot** showing the 4 DHCP DORA packets and the assigned IP/gateway options.

Part 2: NAT Overload (PAT) Header & Translation Table Analysis

2. Compare the **ICMP Echo Request** packet captured on the **Inside LAN link** (`HostA -> R1 Gi0/0`) against the corresponding **ICMP Echo Request** packet captured on the **Outside WAN link** (`R1 Gi0/1 -> Internet`) when pinging `203.0.113.2` :
   - What are the **Source IP**, **Destination IP**, and **Layer 2 Source/Destination MAC addresses** on the Inside LAN link versus the Outside WAN link?
   - Explain how **Port Address Translation (NAT Overload / PAT)** allows dozens of internal hosts on `192.168.1.0/24` to simultaneously access external networks using a single public IPv4 address (`203.0.113.1`). How does the router use Layer 4 TCP/UDP port numbers (or ICMP Query Identifiers) to demultiplex return traffic back to the correct internal host?
   - Reference your `show ip nat translations` CLI output and explain the meaning of the four NAT address terms: **Inside Local**, **Inside Global**, **Outside Local**, and **Outside Global**.
   - Include an **annotated Wireshark screenshot** highlighting the rewritten Source IP (`203.0.113.1`) on the Outside WAN capture.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything]
Discovery Method: [How did you find the issue?]
Resolution: [How did you fix it?]

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9. What to Hand In

Submit the following files to Canvas:

1. **CML Topology File (`.yaml`)**: Exported CML `.yaml` file with your name, date, and lab title in comments at the top of the file.
2. **Wireshark Packet Capture File(s) (`.pcap` ONLY — NOT `.pcapng`)**:
   - Your captured `.pcap` file(s) (`dhcp_nat_traffic.pcap` , or `inside_dhcp.pcap` and `outside_nat.pcap`) containing the DHCP DORA negotiation and NAT-translated ICMP traffic.
3. **Lab Report (`.pdf`)** containing:
   - Screenshot of your running CML topology showing custom node names (`<initials>-HostA` , `<initials>-R1` , `<initials>-Internet`).
   - Router R1 CLI screenshots showing:

    - `show ip dhcp binding` (confirming the active DHCP lease for `<initials>-HostA`).
    - `show ip nat translations` (confirming active NAT translations after pinging `203.0.113.2`).
- Annotated Wireshark screenshots showing:
    - The 4-way DHCP DORA handshake (`Discover`, `Offer`, `Request`, `ACK`) on the Inside link.
    - The NAT-translated ICMP Echo Request/Reply on the Outside link showing the source IP rewritten to `203.0.113.1`.
- Complete written answers to **Part 1 (DHCP Protocol Analysis)** and **Part 2 (NAT Overload / PAT Analysis)** in Section 7, plus Section 8 (**Troubleshooting**).